

Enable CloudTrail monitoring and store the events in S3 and CloudWatch log events.

CloudTrail > Quick trail create

Start logging management events by creating a trail with simplified settings. Logs are sent to an S3 bucket we create on your behalf. To choose a different bucket or additional events, go to the full [Create trail](#) workflow.

A trail created in the console is a multi-region trail. [Learn more](#)

Trail name
Enter a display name for your trail.

my-cloudtrail

3-128 characters. Only letters, numbers, periods, underscores, and dashes are allowed.

Trail log bucket and folder

aws-cloudtrail-logs-469132170253-fd55dc02

Logs will be stored in aws-cloudtrail-logs-469132170253-fd55dc02/AWSLogs/469132170253

Though there is no cost to log these events, you incur charges for the S3 bucket that we create to store your logs.

Cancel Create trail

Create a cloudtrail

Trails

Copy events to Lake Delete Create trail

	Name	Home region	Multi-region trail	ARN	Insights	Organization trail	S3 bucket	Log file prefix
☐	my-cloudtrail	US East (N. Virginia)	Yes	arn:aws:cloudtrail:us-east-1:469132170253:trail/my-cloudtrail	Disabled	No	aws-cloudtrail-logs-469132170253-fd55dc02	-

CloudTrail > Trails > [arn:aws:cloudtrail:us-east-1:469132170253:trail/my-cloudtrail](#) > Edit

CloudWatch Logs Info

☒ Enabled

Log group Info

☒ New

☐ Existing

Log group name

aws-cloudtrail-logs-469132170253-57fea30b

1-512 characters. Only letters, numbers, dashes, underscores, forward slashes, and periods are allowed.

IAM Role Info

AWS CloudTrail assumes this role to send CloudTrail events to your CloudWatch Logs log group.

☒ New

☐ Existing

Role name

CloudTrailRoleForCloudWatchLogs_my-cloudtrail

Use the new log group and IAM role

Enable SNS for CloudTrail to send alerts via email.

✔ Topic Cloudtrail-alerts created successfully.

You can create subscriptions and send messages to them from this topic.

Publish message



Cloudtrail-alerts

Edit

Delete

Publish message

Details

Name

Cloudtrail-alerts

ARN

arn:aws:sns:us-east-1:469132170253:Cloudtrail

Display name

-

Type

Standard

Firstly create the Cloudtrail-alerts

Then create subscription

Subscription: 754ca3d8-7626-4717-b59a-83b7d1055925

Edit

Delete

Details

ARN

arn:aws:sns:us-east-1:469132170253:Cloudtrail-alerts:754ca3d8-7626-4717-b59a-83b7d1055925

Endpoint

botkajagan@gmail.com

Topic

[Cloudtrail-alerts](#)

Status

Pending confirmation

Protocol

EMAIL

Create subscription with your email id

Details

ARN

arn:aws:sns:us-east-1:469132170253:Cloudtrail-alerts:754ca3d8-7626-4717-b59a-83b7d1055925

Endpoint

You get an email regarding confirmation, the confirm

Status

Confirmed

Protocol

EMAIL

SNS notification delivery [Info](#)

☒ Enabled

Create a new SNS topic

☐ New

☒ Existing

SNS topic

arn:aws:sns:us-east-1:469132170253:Cloudtrail-alerts

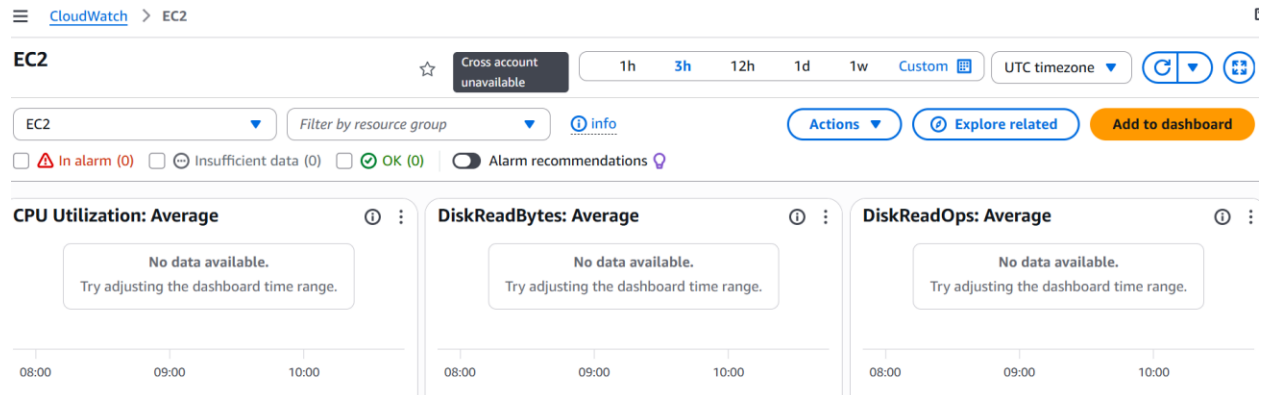


Cancel

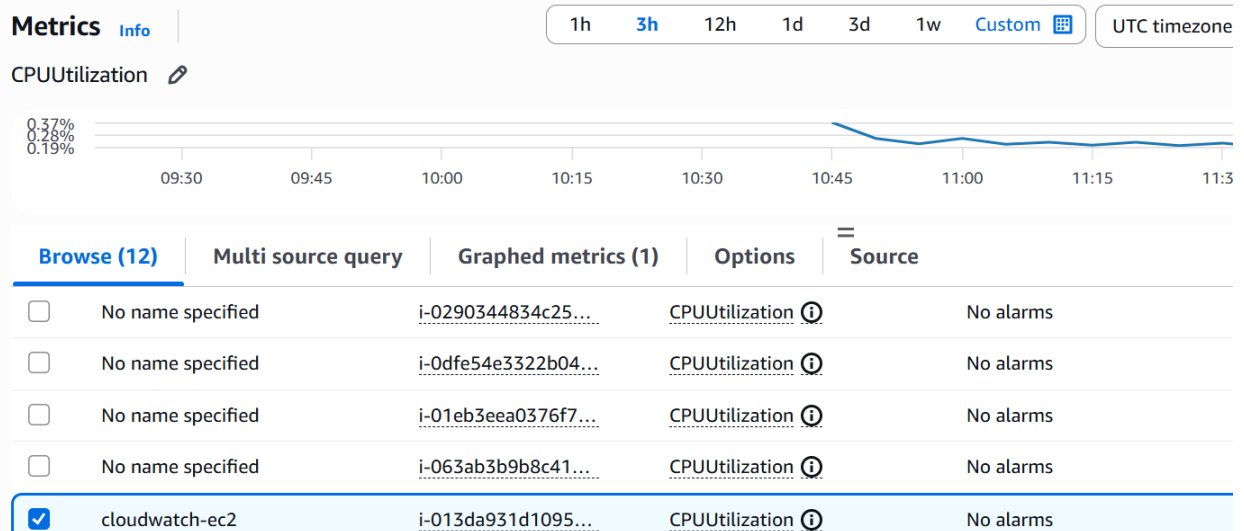
Save changes

Enable the SNS notification, then save changes

Configure CloudWatch monitoring and record the CPU utilization and other metrics of EC2.



From the cloudwatch choose ec2



Create one alarm to send an alert to email if the CPU utilization is more than 70 percent.

Go to Merics – cpuutilization – search for cloudwatch-ec2

Specify metric and conditions

Data source

Select the type of data to monitor

☒ Metrics

Monitor CloudWatch metrics

☐ Logs

Monitor CloudWatch log queries

☐ Alarms

Combine multiple alarms

Type

Choose the metric type for this alarm.

☒ Classic

Use the standard CloudWatch metric selection experience.

☐ PromQL

Use PromQL queries to define the metric for this alarm.

Create alarm select metric

Metric

[Edit](#)

Namespace
AWS/EC2

Metric name
CPUUtilization

InstanceId
i-013da931d1095b509

Instance name
cloudwatch-ec2

Statistic

Q Average X

Period

5 minutes ▼

Select the metric = cloudwatch-ec2

Put threshold value as greater than 70

Alarm state trigger

Define the alarm state that will trigger this action.

[Remove](#)

☒ **In alarm**
The metric or expression is outside of the defined threshold.

☐ **OK**
The metric or expression is within the defined threshold.

☐ **Insufficient data**
The alarm has just started or not enough data is available.

Send a notification to the following SNS topic

Define the SNS (Simple Notification Service) topic that will receive the notification.

- ☒ Select an existing SNS topic
- ☐ Create new topic
- ☐ Use topic ARN to notify other accounts

Send a notification to...

Q cloudwatch-alerts X

Only topics belonging to this account are listed here. All persons and applications subscribed to the selected topic will receive notifications.

Email (endpoints)

botkajagan@gmail.com - [View in SNS Console](#)

Add your email

Some subscriptions are pending confirmation
Amazon SNS doesn't send messages to an endpoint until the subscription is confirmed

View SNS Subscriptions



Alarms

Mute rules

Alarms (1)

Clear selection



Create composite alarm

Actions

Create alarm

Filter alarms

Quick filters

< 1 > ⚙

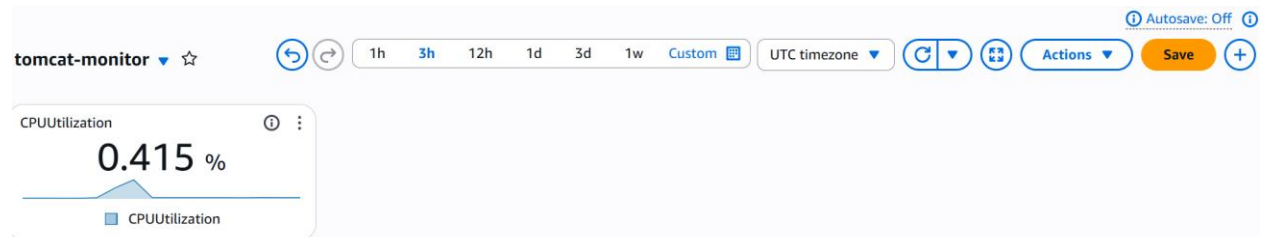
☐	Name	State	Actions	Actions muted	Last state up
☐	High-CPU-Cloudwatch-EC2	OK	Actions enabled Warning	-	2026-09-09

Alarm has successfully set

Create a Dashboard and monitor the Tomcat service whether it is running or not and send the alert.

```
Complete!
[root@ip-10-0-0-25 bin]# java -version
openjdk version "11.0.32.1" 2026-08-18 LTS
OpenJDK Runtime Environment Corretto-11.0.32.10.1 (build 11.0.32.1+10-LTS)
OpenJDK 64-Bit Server VM Corretto-11.0.32.10.1 (build 11.0.32.1+10-LTS, mixed mode)
[root@ip-10-0-0-25 bin]# ./startup.sh
Using CATALINA_BASE:   /opt/apache-tomcat-9.0.121
Using CATALINA_HOME:   /opt/apache-tomcat-9.0.121
Using CATALINA_TMPDIR: /opt/apache-tomcat-9.0.121/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /opt/apache-tomcat-9.0.121/bin/bootstrap.jar:/opt/apache-tomcat-9.0.121/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
```

Create Dashboard in the cloudwatch



Add Cpu utilization of tomcat

Browse (22)	Multi source query	Graphed metrics (1)	Options	Source
InstancedId = i-0424af31afc345ed0 Clear filters				
☐	Instance name 22/22	InstancedId	Metric name	Alarms
☒	Amazon linux	i-0424af31afc345...	CPUUtilization	No alarms
☐	Amazon linux	i-0424af31afc345...	InstanceEBSIOPSExceededCheck	No alarms
☐	Amazon linux	i-0424af31afc345...	InstanceERSThroughputExceededCheck	No alarms
				Cancel Select metric

> Create alarm

Preview

1h 3h 12h 1d 3d 1w Custom UTC timezone

Graph

This alarm will trigger when the blue line goes above the red line for 1 datapoints within 5 minutes.

Percent

6.05%
3.13%
0.2%

15:30 16:00 16:30 17:00 17:30 18:00

— CPUUtilization

☐ Greater/Equal
>= threshold

☐ Lower/Equal
<= threshold

☐ Lower
< threshold

than...
Define the threshold value.

70

Must be a number.

► Additional configuration

Alarms (2)

Clear selection Create composite alarm Actions Create alarm

Filter alarms Quick filters

☐	Name	State	Actions	Actions muted	Last state up
☐	tomcat-CPU-alert	Insufficient data	Actions enabled Warning	-	2026-09-09

Alarm has been successfully configured

For confirmation Im changing threshold to 0

Create a Dashboard and monitor the Nginx service to send the alert if Nginx is not running.

```
[root@ip-10-0-0-25 ~]# systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; disabled; preset: disabled)
   Active: active (running) since Wed 2026-09-09 18:30:17 UTC; 3s ago
     Process: 29958 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)
     Process: 29959 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)
     Process: 29960 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)
   Main PID: 29961 (nginx)
    Tasks: 3 (limit: 1059)
   Memory: 4.1M
      CPU: 59ms
   CGroup: /system.slice/nginx.service
           └─29961 "nginx: master process /usr/sbin/nginx"
             └─29962 "nginx: worker process"
               └─29963 "nginx: worker process"

Sep 09 18:30:17 ip-10-0-0-25.ec2.internal systemd[1]: Starting nginx.service - The nginx HTTP and reverse proxy server...
Sep 09 18:30:17 ip-10-0-0-25.ec2.internal nginx[29959]: nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
Sep 09 18:30:17 ip-10-0-0-25.ec2.internal nginx[29959]: nginx: configuration file /etc/nginx/nginx.conf test is successful
Sep 09 18:30:17 ip-10-0-0-25.ec2.internal systemd[1]: Started nginx.service - The nginx HTTP and reverse proxy server.
[root@ip-10-0-0-25 ~]#
```

Nginx is running

Nginx-monitoring ☆

1h 3h 12h 1d 3d 1w Custom UTC timezone Actions Save +

CPUUtilization

0.39 %

CPUUtilization

Created a Nginx-monitoring Dashboard for Nginx

```
[root@ip-10-0-0-25 ~]# dnf install -y amazon-cloudwatch-agent
Last metadata expiration check: 1:46:51 ago on Wed Sep  9 17:04:36 2026.
Dependencies resolved.
=====
Package                                Architecture    Version                               Repository    Size
=====
Installing:
amazon-cloudwatch-agent                x86_64          1.300069.1-1.amzn2023                amazonlinux   69
Transaction Summary
=====
Install 1 Package
```

We need to install cloudwatch agent

```
[root@ip-10-0-0-25 ~]# /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-config-wizard
Starting config-wizard, this will map back to a call to amazon-cloudwatch-agent
Executing /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent with arguments: [config-wizard]=====
= Welcome to the Amazon CloudWatch Agent Configuration Manager =
=
= CloudWatch Agent allows you to collect metrics and logs from =
= your host and send them to CloudWatch. Additional CloudWatch =
= charges may apply. =
=====
On which OS are you planning to use the agent?
1. linux
2. windows
3. darwin
default choice: [1]:
1
```

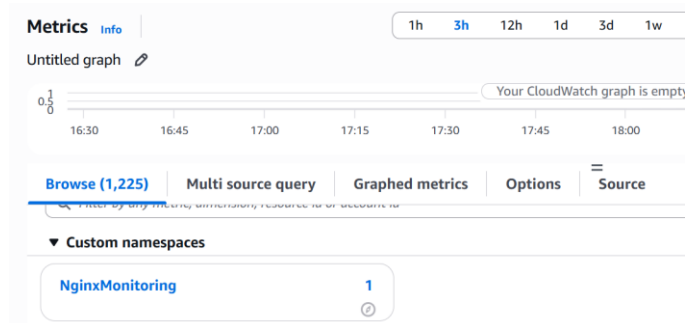
```
=====
On which OS are you planning to use the agent?
1. linux
2. windows
3. darwin
default choice: [1]:
1
Trying to fetch the default region based on ec2 metadata...
1: imds retry client will retry 1 timesAre you using EC2 or On-Premises hosts?
1. EC2
2. On-Premises
default choice: [1]:
1
Which user are you planning to run the agent?
1. cwagent
2. root
3. others
default choice: [1]:
1
Do you want to turn on StatsD daemon?
1. yes
2. no
default choice: [1]:
2
Do you want to monitor metrics from Collectd? WARNING: Collectd must be installed or the Agent will fail to start
1. yes
2. no
default choice: [1]:
2
Do you want to monitor any host metrics? e.g. CPU, memory, etc.
1. yes
2. no
default choice: [1]:
2
Do you have any existing CloudWatch Log Agent (http://docs.aws.amazon.com/AmazonCloudWatch/latest/logs/AgentReference.html) configuration file to import for migration?
1. yes
2. no
default choice: [2]:
2
Do you want to monitor any log files?
1. yes
2. no
default choice: [1]:
1
Do you want the CloudWatch agent to also retrieve X-ray traces?
1. yes
```

```
[root@ip-10-0-0-25 ~]# cat > /opt/aws/amazon-cloudwatch-agent/bin/config.json <<'EOF'
{
  "agent": {
    "run_as_user": "cwagent"
  },
  "metrics": {
    "namespace": "NginxMonitoring",
    "metrics_collected": {
      "procstat": [
        {
          "pattern": "nginx",
          "measurement": [
            "pid_count"
          ],
          "metrics_collection_interval": 60
        }
      ]
    }
  }
}
EOF
[root@ip-10-0-0-25 ~]# |
```

Runthis prompt

```
[root@ip-10-0-0-25 ~]# sudo systemctl status amazon-cloudwatch-agent
● amazon-cloudwatch-agent.service - Amazon CloudWatch Agent
   Loaded: loaded (/etc/systemd/system/amazon-cloudwatch-agent.service; enabled; preset: disabled)
   Active: active (running) since wed 2026-09-09 19:24:39 UTC; 43s ago
     Main PID: 32431 (amazon-cloudwat)
        Tasks: 7 (limit: 1059)
       Memory: 47.6M
          CPU: 404ms
     CGroup: /system.slice/amazon-cloudwatch-agent.service
             └─32431 /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent -config /opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwatch-agent.toml

Sep 09 19:24:40 ip-10-0-0-25.ec2.internal start-amazon-cloudwatch-agent[32436]: Executing /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent with
Sep 09 19:24:40 ip-10-0-0-25.ec2.internal start-amazon-cloudwatch-agent[32441]: D! [EC2] Found active network interface
Sep 09 19:24:40 ip-10-0-0-25.ec2.internal start-amazon-cloudwatch-agent[32441]: I! imds retry client will retry 1 times! Detected the instance is EC2
Sep 09 19:24:40 ip-10-0-0-25.ec2.internal start-amazon-cloudwatch-agent[32441]: 2026/09/09 19:24:40 Reading json config file path: /opt/aws/amazon-cloudwat
Sep 09 19:24:40 ip-10-0-0-25.ec2.internal start-amazon-cloudwatch-agent[32441]: /opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwatch-agent.json does not
Sep 09 19:24:40 ip-10-0-0-25.ec2.internal start-amazon-cloudwatch-agent[32441]: 2026/09/09 19:24:40 Reading json config file path: /opt/aws/amazon-cloudwat
```



Notification

Alarm state trigger

Define the alarm state that will trigger this action.

Remove



In alarm

The metric or expression is outside of the defined threshold.



OK

The metric or expression is within the defined threshold.



Insufficient data

The alarm has just started or not enough data is available.

Send a notification to the following SNS topic

Define the SNS (Simple Notification Service) topic that will receive the notification.



Select an existing SNS topic



Create new topic



Use topic ARN to notify other accounts

Send a notification to...



cloudwatch-alerts



Only topics belonging to this account are listed here. All persons and applications subscribed to the selected topic will receive notifications.

Alarms (3)							Clear selection		Create composite alarm	Actions ▼	Create alarm
Filter alarms							Quick filters				
☐	Name ▼	State ▼	Actions ▼	Actions muted ▼	Last state update						
☐	Nginx Alarm	Insufficient data	Actions enabled Warning	-	2026-09-09 19:3:						
☐	tomcat-CPU-alert	In alarm	Actions enabled Warning	-	2026-09-09 18:11						
☐	High-CPU-Cloudwatch-EC2	Insufficient data	Actions disabled permanently Warning	-	2026-09-09 16:41						

Alarm has been set for Nginx

```
Sep 09 18:30:17 ip-10-0-0-25.ec2.internal systemd[1]: Started nginx.service - The nginx HTTP and reverse proxy server.
[root@ip-10-0-0-25 ~]# systemctl stop nginx
[root@ip-10-0-0-25 ~]# systemctl status nginx
○ nginx.service - The nginx HTTP and reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; disabled; preset: disabled)
   Active: inactive (dead)

Sep 09 18:30:17 ip-10-0-0-25.ec2.internal systemd[1]: Starting nginx.service - The nginx HTTP and reverse proxy server...
Sep 09 18:30:17 ip-10-0-0-25.ec2.internal nginx[29959]: nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
Sep 09 18:30:17 ip-10-0-0-25.ec2.internal nginx[29959]: nginx: configuration file /etc/nginx/nginx.conf test is successful
Sep 09 18:30:17 ip-10-0-0-25.ec2.internal systemd[1]: Started nginx.service - The nginx HTTP and reverse proxy server.
Sep 09 19:35:45 ip-10-0-0-25.ec2.internal systemd[1]: Stopping nginx.service - The nginx HTTP and reverse proxy server...
Sep 09 19:35:45 ip-10-0-0-25.ec2.internal systemd[1]: nginx.service: Deactivated successfully.
Sep 09 19:35:45 ip-10-0-0-25.ec2.internal systemd[1]: Stopped nginx.service - The nginx HTTP and reverse proxy server.
[root@ip-10-0-0-25 ~]# |
```

Stop the alarm and check for the result

☐ [Nginx Alarm](#)

 In alarm

 Actions enabled [Warning](#)

-

2026-09-09 19:40